DISASTERLINK – INTELLIGENT DISASTER RELIEF AND RESOURCE COORDINATION PLATFORM USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

AADHAVAN RP

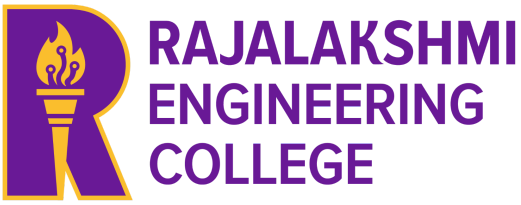
(240701819)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai) Rajalakshmi Nagar,

Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "DisasterLink – Intelligent Disaster Relief and Resource Coordination Platform using Zoho Creator" is the bonafide work of AADHAVAN RP 240701819, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

SIGNATURE

Dr. MANORANJINI.J
HEAD OF THE DEPARTMENT
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

DR J JINU SOPHIA
COURSE FACULTY
Associate Professor
Department of Computer Science and Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on

_____

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, AADHAVAN RP (240701819), hereby declare that the project report titled "DisasterLink – Intelligent Disaster Relief and Resource Coordination Platform using Zoho Creator", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the project work carried out by me.

I further declare that the work presented in this report, including the application design, forms, lookup relationships, reports, dashboard, Blueprint configuration and Deluge workflows, has been developed as part of this project. This work has not been submitted, either in part or in full, for the award of any other degree or course credit.

I have acknowledged the relevant references and resources used in preparing this project report.

Place: Chennai

Date: 11.11.2026

SIGNATURE OF THE STUDENT

AADHAVAN RP

240701819

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman, Mr. S. Meganathan, B.E., F.I.E., Chairperson, Dr. Mrs. Thangam Meganathan, M.A., M.Phil., Ph.D., for providing the infrastructure and facilities that made this work possible.

I am grateful to our Dr. S.N. Murugesan, M.E., Ph.D., Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. Manoranjini.J, Head of the Department, Department of Computer Science and Engineering, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty DR. Jinu Sophia for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge [Zoho Corporation / Zoho Creator Education Programme] for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

AADHAVAN RP
240701819

ABSTRACT

Disaster management is a critical process that requires effective coordination, timely communication and efficient distribution of relief resources during emergencies such as floods, cyclones, earthquakes and fires. Traditional disaster relief operations often rely on manual records, phone calls and disconnected systems, which can lead to delays in responding to emergency requests, inefficient resource allocation and difficulties in tracking available relief materials.

DisasterLink – Intelligent Disaster Relief and Resource Coordination Platform is a low-code application developed using **Zoho Creator** to simplify and organise disaster relief operations. The system provides a centralised platform for managing disaster incidents, emergency requests, relief inventory, responder information and response assignments.

The application consists of five core forms: **Disaster Incident, Emergency Request, Relief Inventory, Responder Profile and Response Assignment**. These forms are connected through lookup relationships to maintain data consistency and improve coordination. Zoho Creator reports and a dashboard support the monitoring of incidents, emergency requests, resource availability and response activities.

The system incorporates **Zoho Creator Blueprint and Deluge scripting** to automate important operations. The Emergency Response Workflow Blueprint supports the approval and assignment process. Deluge workflows validate inventory allocations, prevent the allocation of resources beyond available stock, update inventory quantities and automate changes to linked emergency request statuses. During implementation testing, the allocation of five Water Filters successfully reduced the available quantity from 250 to 245 and increased the reserved quantity from 50 to 55.

The project demonstrates how a low-code platform can support disaster relief coordination by reducing manual work, improving resource tracking and organising emergency response activities. Future enhancements may include citizen portal access, real-time notifications, geographical mapping, AI-based disaster prediction and advanced analytics.

Keywords: *Disaster Management, DisasterLink, Zoho Creator, Low-Code Application Development, Emergency Response, Relief Inventory Management, Deluge Scripting, Blueprint, Workflow Automation, Resource Coordination.*

PROJECT AT A GLANCE

Attribute	Details
Project Title	DisasterLink – Intelligent Disaster Relief and Resource Coordination Platform
Application Name	DisasterLink Emergency System
Application Domain	Disaster Management / Emergency Relief Coordination
Student Name	Tanuj Krrishna.A
Register Number	To be entered
Institution	Rajalakshmi Engineering College
Development Platform	Zoho Creator
Core Modules	Disaster Incident, Emergency Request, Relief Inventory, Responder Profile, Response Assignment
Dashboard	Assignment, request, inventory and incident summary; priority and inventory charts
Automation	Deluge validation and workflows; Emergency Response Workflow Blueprint
Reports	Response Assignment Report and module-wise operational reports
Deployment URL	To be added after publishing and verification
Academic Year	2026–2027

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update and Delete
DS	Deluge Script
ER	Entity–Relationship
ID	Identifier
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UI	User Interface
URL	Uniform Resource Locator
WF	Workflow
Zoho Creator	Cloud-based low-code application development platform

CHAPTER 1
INTRODUCTION

1.1 Background

Natural disasters and emergency situations create urgent requirements for rescue services, food, drinking water, shelter, medical assistance and other essential materials. Effective relief operations depend on accurate information about the incident, the people requesting help, the resources available and the responders assigned to deliver support.

DisasterLink is designed as a centralized disaster relief and resource coordination application. It brings together disaster incidents, emergency requests, relief inventory, responder profiles and response assignments. Zoho Creator is used as the low-code development platform because it supports visual forms, reports, pages, workflows and Deluge scripting.

1.2 Problem Statement

When emergency information is managed through separate registers, spreadsheets or messaging channels, relief teams may face delays and inconsistent records. Coordinators may not have a consolidated view of pending requests, available stock, assigned responders and delivery progress.

- Emergency requests may be overlooked or duplicated when they are recorded in separate channels.
- Inventory quantities may become inaccurate when reserved and available stock are not updated together.
- Responders may not receive clear assignment details, destination information or delivery instructions.
- Priority levels may not be visible to coordinators when multiple requests arrive at the same time.
- Management may lack a consolidated dashboard to review incidents, requests and relief operations.

1.3 Objectives

- To maintain disaster incident details in a centralized application.
- To record emergency requests with requester details, request type, priority and lives at risk.
- To maintain relief inventory and monitor available and reserved quantities.
- To maintain responder profiles for emergency coordination.
- To create response assignments linking requests, responders and relief resources.
- To automate inventory validation and quantity updates using Deluge workflows.
- To provide reports and dashboard views for operational monitoring.

1.4 Scope of the Project

The project covers the five principal modules: Disaster Incident, Emergency Request, Relief Inventory, Responder Profile and Response Assignment. It includes data capture, lookup relationships, operational reports, dashboard indicators, priority categorization and workflow-based inventory coordination. The supplied screenshots demonstrate the dashboard, Emergency Request form, Response Assignment form and Response Assignment Report.

Advanced geographic routing, SMS gateway integration, citizen self-service portals, predictive disaster analytics and integrations with government emergency systems are considered future enhancements unless separately configured and verified.

1.5 Justification for a Low-Code Approach

The application is data-centric and workflow-oriented. Zoho Creator provides visual form and report builders, lookup fields for connecting records, dashboard pages for summaries and Deluge scripting for business rules.

This approach reduces the amount of conventional code required while allowing project-specific validation and automation.

1.6 Organisation of the Report

Chapter 2 studies the existing process and the proposed system. Chapter 3 records stakeholders and requirements. Chapter 4 presents architecture, data model, forms and workflows. Chapter 5 describes the Zoho Creator implementation. Chapter 6 presents the testing approach and test cases. Chapter 7 discusses deployment and security. Chapter 8 reviews results and limitations. Chapter 9 concludes the report and lists future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a manual disaster relief process, incident details may be recorded by one team, emergency requests collected through phone calls or paper forms, and relief stock maintained in separate inventory registers. Responder assignments may be communicated through calls or messaging groups. Delivery confirmation and remaining stock may then be updated manually.

2.2 Limitations of the Existing System

- Information is distributed across different records and communication channels.
- It may be difficult to identify the current status and priority of every request.
- Manual stock updates can cause differences between physical and recorded quantities.
- Assignment details and delivery evidence may not be stored in a consistent format.
- Preparing operational summaries may require repeated manual compilation.

2.3 Low-Code versus Traditional Development

Criterion	Traditional Development	Low-Code using Zoho Creator
Development	Separate UI, database and service coding	Visual forms, reports and workflows
Skills	Programming and database development	Platform configuration and Deluge scripting
Data capture	Custom forms and database operations	Creator form builder and lookup fields
Automation	Custom code and job scheduling	Creator workflows and Deluge
Maintenance	Code and infrastructure maintenance	Visual configuration and workflow maintenance
Flexibility	Broad control over implementation	Bound by platform features and plan limits

2.4 Literature and Platform Study

The project applies model-driven low-code development, structured data capture, relational lookups, workflow automation and role-oriented reporting. Zoho Creator documentation and Deluge language references are relevant resources for implementing forms, reports, pages, permissions and record operations. The exact features available depend on the account edition and configuration.

2.5 Proposed System

The proposed DisasterLink application stores incident, request, inventory, responder and assignment records in one environment. Emergency requests can be prioritized, relief inventory can be reviewed before allocation, and response assignments can record the responder and delivery information. Dashboard cards and reports provide a consolidated view of the operational records.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Role	Description	Responsibilities
Administrator	Application administrator	Manage modules, users, master records, reports and settings.
Relief Coordinator	Coordinates relief activities	Review requests, assess priority, check inventory and create assignments.
Responder	Person/team assigned to a response	Receive assignment details, dispatch resources and update delivery information.
Inventory Manager	Maintains relief stock	Add inventory records and monitor available/reserved quantities.
Management	Operational oversight	Review dashboard indicators, reports and incident summaries.

3.2 Functional Requirements

ID	Functional Requirement	Priority
FR-01	The system shall record disaster incidents with a unique reference and incident details.	High
FR-02	The system shall record emergency requests with requester details and request type.	High
FR-03	The system shall capture priority level and lives at risk for each emergency request.	High
FR-04	The system shall maintain relief inventory categories and quantities.	High
FR-05	The system shall maintain responder profile information.	High
FR-06	The system shall create response assignments linked to relevant request and responder records.	High
FR-07	The system shall validate that allocated quantity is positive and does not exceed available stock.	High
FR-08	The system shall update available and reserved quantities after successful allocation.	High
FR-09	The system shall record dispatch time, delivery time and completion notes.	Medium
FR-10	The system shall provide reports and dashboard summaries for operational monitoring.	Medium

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	Forms should use clear labels and understandable validation messages.
NFR-02	Data Integrity	Inventory allocation must not exceed available quantity.
NFR-03	Security	Only authorized users should access or modify operational records.

NFR-04	Availability	The application should be accessible when Zoho Creator and network services are available.
NFR-05	Maintainability	Forms, fields and workflows should use consistent names and documented logic.
NFR-06	Performance	Reports and dashboards should display relevant summaries without unnecessary data retrieval.

3.4 Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator cloud-based low-code platform
Scripting	Deluge
Browser	Supported modern browser such as Chrome, Edge or Firefox
Client Device	Desktop or laptop; tablet/mobile where page design supports it
Connectivity	Internet connection
Storage	Zoho Creator application data storage
Evidence	Screenshots of dashboard, forms and reports

3.5 Use Case Model

The main actors are Administrator, Relief Coordinator and Responder. The administrator maintains records and application configuration. The relief coordinator reviews emergency requests, checks relief inventory and creates response assignments. Responders use assignment details to carry out dispatch and delivery updates.

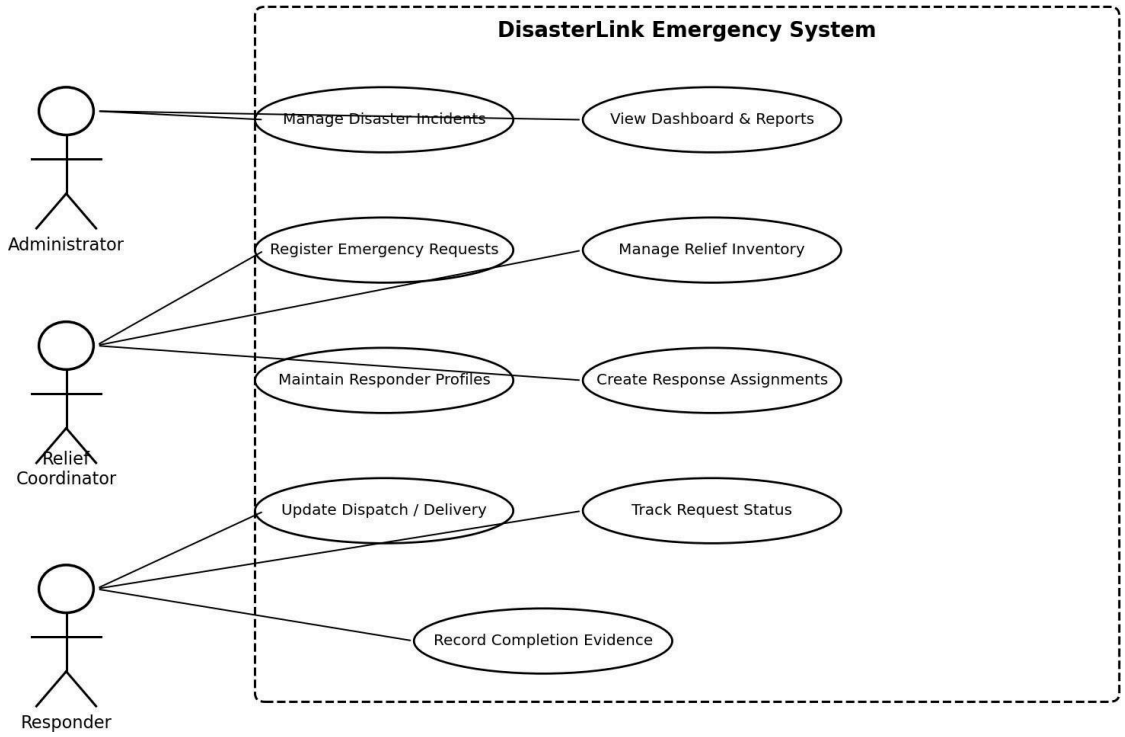


Figure 3.1: Use Case Diagram

Attribute	Description

Use Case ID	UC-01
Use Case Name	Create and process an emergency response assignment
Actors	Relief Coordinator, Responder, Administrator
Pre-condition	An emergency request, responder profile and relevant inventory record are available.
Main Flow	Coordinator reviews request; checks priority and stock; selects responder and destination; enters assignment and dispatch details; system validates quantity; assignment is saved and inventory is updated.
Alternate Flow	If the quantity is invalid or exceeds available stock, the system rejects the allocation and displays a validation message.
Post-condition	Assignment record is stored and inventory quantities reflect the allocation.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

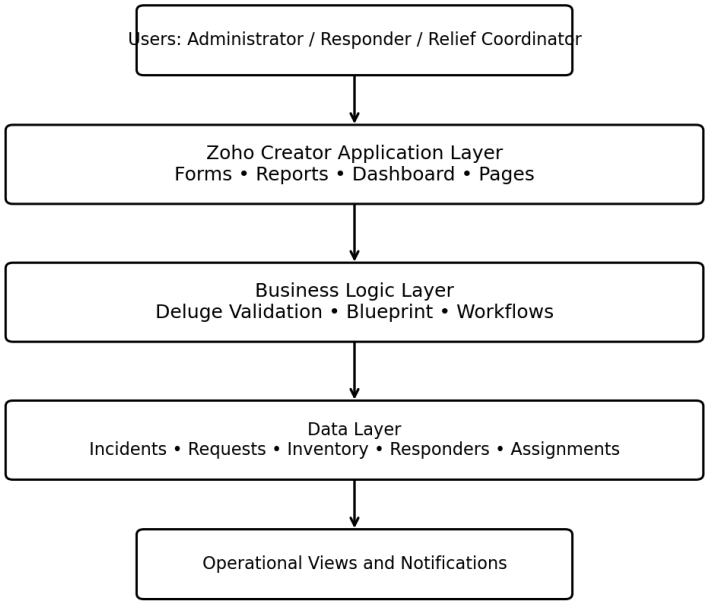


Figure 4.1: Proposed System Architecture

The architecture contains a user layer, Zoho Creator application layer, business logic layer, data layer and operational output layer. Users access forms, reports and dashboards through the application. Deluge validation and workflows apply business rules. The data layer stores incident, request, inventory, responder and assignment records. Reports and dashboard components present operational information.

4.2 Data Model

Entity / Form	Key Fields	Relationship
Disaster Incident	Incident Reference, Incident Type, Location, Description, Date	Can be associated with multiple emergency requests.
Emergency Request	Request Reference, Request Date, Requester, Phone, Address, Type, Priority, Lives at Risk	Linked to incident and can be used in response assignment.
Relief Inventory	Inventory Reference, Category, Item Name, Available Quantity, Reserved Quantity	Inventory items are allocated through assignments.
Responder Profile	Responder ID, Name, Phone, Responder Type, Availability	One responder can receive multiple assignments over time.
Response Assignment	Assignment Reference, Date, Responder, Phone, Destination, Dispatch Time, Delivery Time, Evidence URL, Notes	Connects response activity to request, responder and allocated resources.

Lookup relationships are used to connect records rather than repeatedly entering the same information. The exact field link names should match the configured Zoho Creator application.

4.3 Form Design

4.3.1 Disaster Incident Form

The Disaster Incident form records the incident reference, incident type, incident date, affected location, description and related information. It provides the incident record to which emergency requests can be associated.

4.3.2 Emergency Request Form

The supplied Emergency Request screenshot shows fields for Request Reference, Request Date, Requester Name, Requester Phone, Requester Address, Request Type, Priority Level, Lives At Risk, Medical Assistance Needed and Description. These fields capture the basic information required to understand the request and determine its urgency.

Field	Purpose
Request Reference	Unique request identifier.
Request Date	Date and time when the request is recorded.
Requester Name / Phone	Contact details of the person requesting assistance.
Requester Address	Location requiring assistance.
Request Type	Category of assistance requested.
Priority Level	Urgency classification.
Lives At Risk	Number of people potentially affected.
Medical Assistance Needed	Indicates whether medical support is required.
Description	Additional details about the emergency.

4.3.3 Relief Inventory Form

The Relief Inventory form maintains the category, item name, total quantity, available quantity and reserved quantity of relief materials. The dashboard screenshot shows categories such as Water, Shelter and Food and a reserved quantity versus available quantity chart.

4.3.4 Responder Profile Form

The Responder Profile form maintains responder identification and contact details, responder type, availability and other coordination information. These records can be selected when creating a response assignment.

4.3.5 Response Assignment Form

The supplied Response Assignment screenshot shows Assignment Reference, Assignment Date, Assigned Responder, Responder Phone, Destination Address, Dispatch Time, Delivery Time, Delivery Evidence URL and Completion Notes. The form captures who is assigned, where the response is directed and the delivery outcome.

Field	Purpose
Assignment Reference	Unique reference for the assignment.
Assignment Date	Date and time of assignment.
Assigned Responder	Responder responsible for the task.
Responder Phone	Contact number for coordination.
Destination Address	Delivery or response location.
Dispatch Time	Time at which resources are dispatched.
Delivery Time	Time at which delivery is completed.
Delivery Evidence URL	Link to supporting delivery evidence, if available.

Completion Notes	Remarks about the completed response.

4.4 Process and Workflow Design

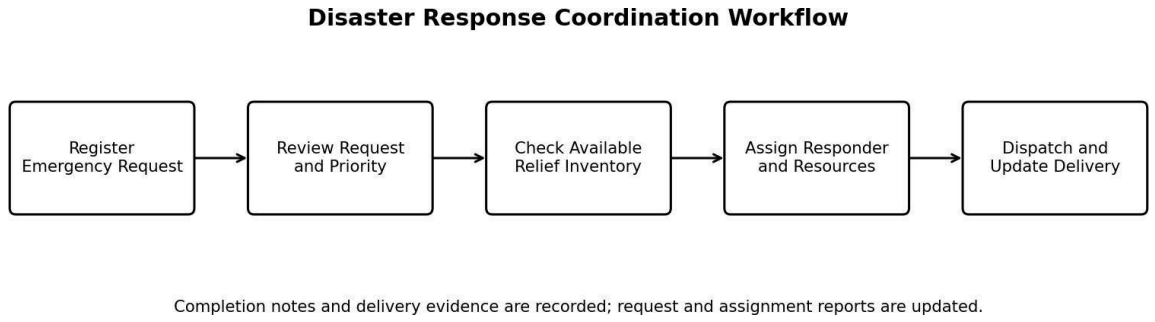


Figure 4.2: Emergency Response Coordination Workflow

The response workflow begins when an emergency request is registered. A coordinator reviews the request and its priority, checks available relief inventory, selects a responder and creates an assignment. Before saving the allocation, the system validates the quantity. Following a successful assignment, available quantity decreases and reserved quantity increases. The assignment can then be updated with dispatch and delivery details, evidence and completion notes.

4.5 Dashboard and Navigation Design

The dashboard screenshot shows four summary cards: Count of Response Assignment (6), Total Emergency Requests (4), Total Relief Inventories (5) and Total Disaster Incidents (1). The Priority Level panel displays High (2), Low (1), Medium (1) and Critical (0). A Reserved Quantity by Available Quantity chart and a Category by Available Quantity chart provide inventory visibility. Quick links are available to add a response assignment, emergency request, relief inventory and disaster incident.

4.6 Access Control Design

Module / Action	Administrator	Relief Coordinator	Responder	Inventory Manager
Manage incident records	Full	Create / View / Update	View assigned context	View
Manage emergency requests	Full	Review / Update	View assigned request	View
Manage inventory	Full	View / Allocate	View allocation	Create / Update
Manage responder profiles	Full	View / Select	View own profile	View
Response assignments	Full	Create / Update	View / Update assigned	View
Dashboard and reports	Full	Operational view	Assigned work view	Inventory view

Permissions must be configured using Creator users, roles, profiles and record-level criteria as supported by the selected account edition. Hiding a menu item alone is not sufficient to protect records.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The DisasterLink Emergency System is created in Zoho Creator. The application contains navigation modules for Dashboard, Response Assignment, Emergency Requests, Relief Inventories, Disaster Incidents and Responder Profiles. The left navigation provides access to module forms and reports.

5.2 Forms and Data Capture

Forms are used to capture incident, request, inventory, responder and assignment data. Lookup fields connect related records. Required fields and field types should be configured to reduce incomplete or invalid entries.

5.3 Reports and Dashboard

The Response Assignment Report screenshot displays assignment references and corresponding date, responder, phone number, destination address and dispatch time. The report contains sample rows with references REF-2026-001 through REF-2026-005 and shows six records in the report view. The report supports review of recorded assignments.

5.4 Deluge Scripting and Validation

Deluge scripting is used to implement business rules in Zoho Creator. The inventory validation described for this project checks that an allocation quantity is greater than zero and does not exceed the available quantity. The validation should run before the assignment is committed.

Illustrative validation logic (adapt the field link names to the actual Creator application):

```
if(input.Allocated_Quantity <= 0)
{
    alert "Allocation quantity must be greater than zero.";
    cancel submit;
}
if(input.Allocated_Quantity > input.Inventory.Available_Quantity)
{
    alert "Allocated quantity exceeds available
    stock."; cancel submit;
}
```

After a valid assignment, the inventory workflow updates the available quantity by subtracting the allocated quantity and updates the reserved quantity by adding the same quantity. The workflow should prevent duplicate execution and retain a reliable record of the transaction.

5.5 Blueprint and Automation

The Emergency Response Workflow Blueprint supports the request approval and assignment process. A request can be reviewed, prioritized and moved through the configured stages. A related workflow updates the emergency request status to Assigned after a successful response assignment. Exact stage names and transitions should match the live Blueprint configuration.

5.6 Integrations

The supplied project description does not establish that external SMS gateways, map services, payment services or government emergency APIs are integrated. Such integrations are therefore treated as future enhancements. The Delivery Evidence URL field can store a reference to evidence hosted elsewhere, subject to the application's access and privacy settings.

5.7 User Management and Role-Based Access

Application users should be assigned roles according to their responsibilities. Administrators manage configuration, relief coordinators process requests and assignments, responders update their assigned work, and inventory managers maintain stock information. Test each role using separate accounts before deployment.

5.8 AI, Analytics and Pro-Code Extension

The current dashboard provides summary cards and charts for operational monitoring. Predictive analytics, AI-based disaster forecasting and custom API integrations are not claimed as implemented in the supplied evidence. They may be evaluated as separate extensions after privacy, accuracy and security requirements are defined.

5.9 Application Screenshots

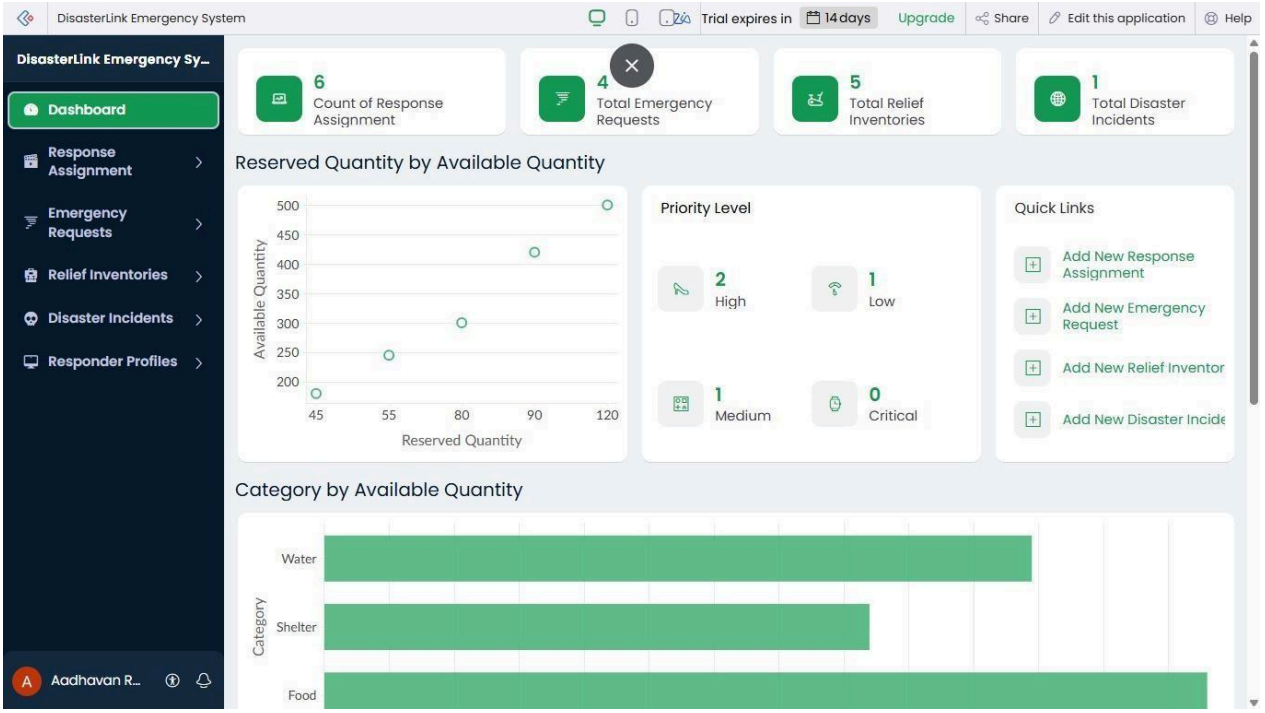


Figure 5.1: DisasterLink Dashboard – summary cards, priority levels, inventory charts and quick links.

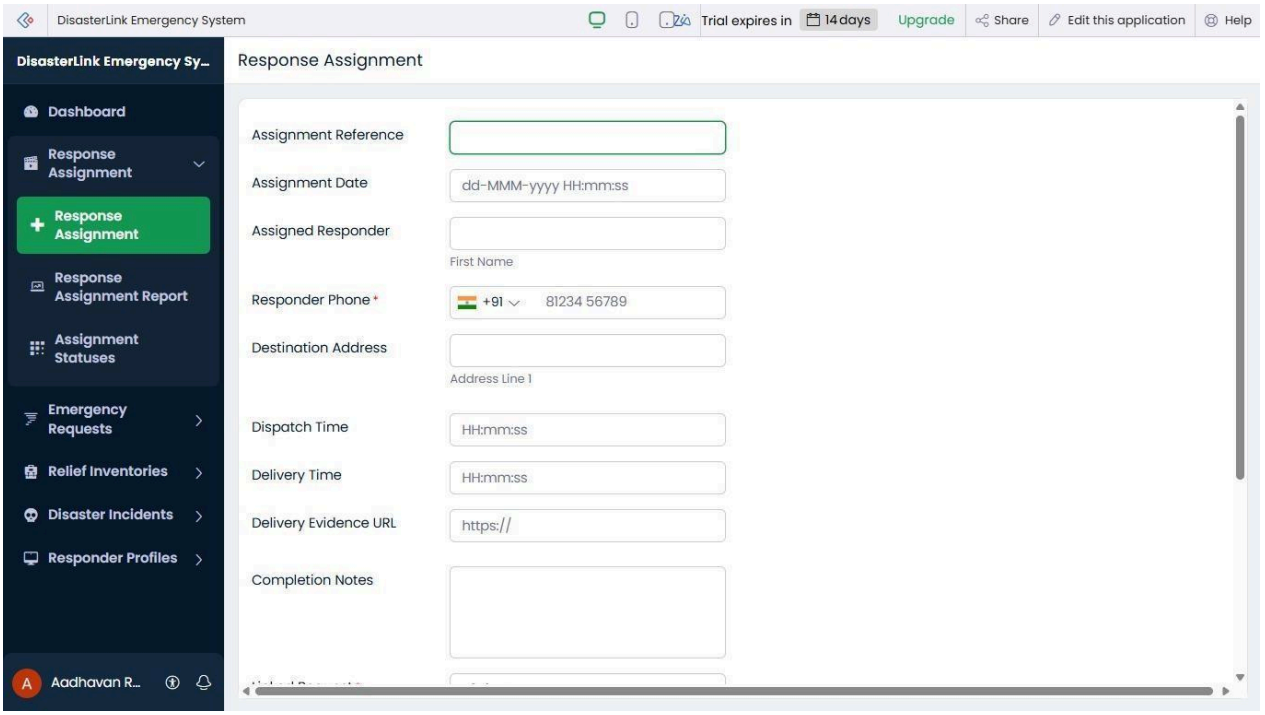


Figure 5.2: Response Assignment form – responder, destination, dispatch and delivery details.

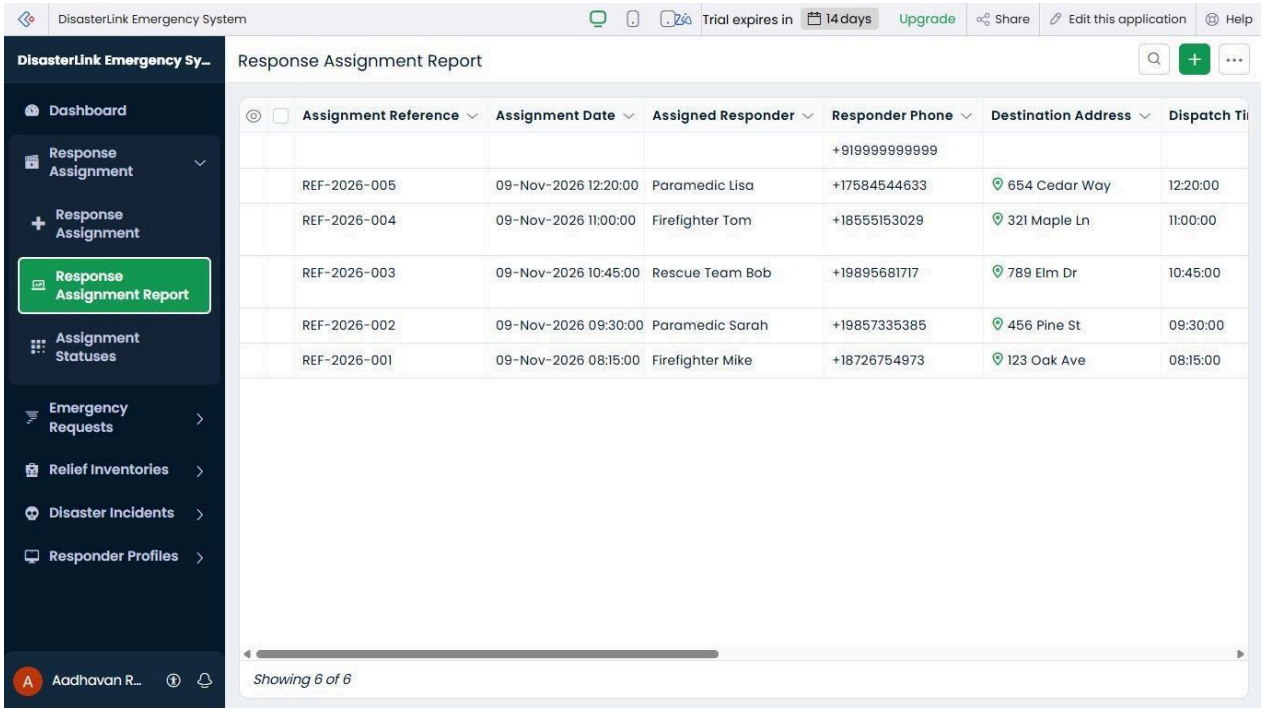


Figure 5.3: Response Assignment Report – assignment reference, date, responder and destination.

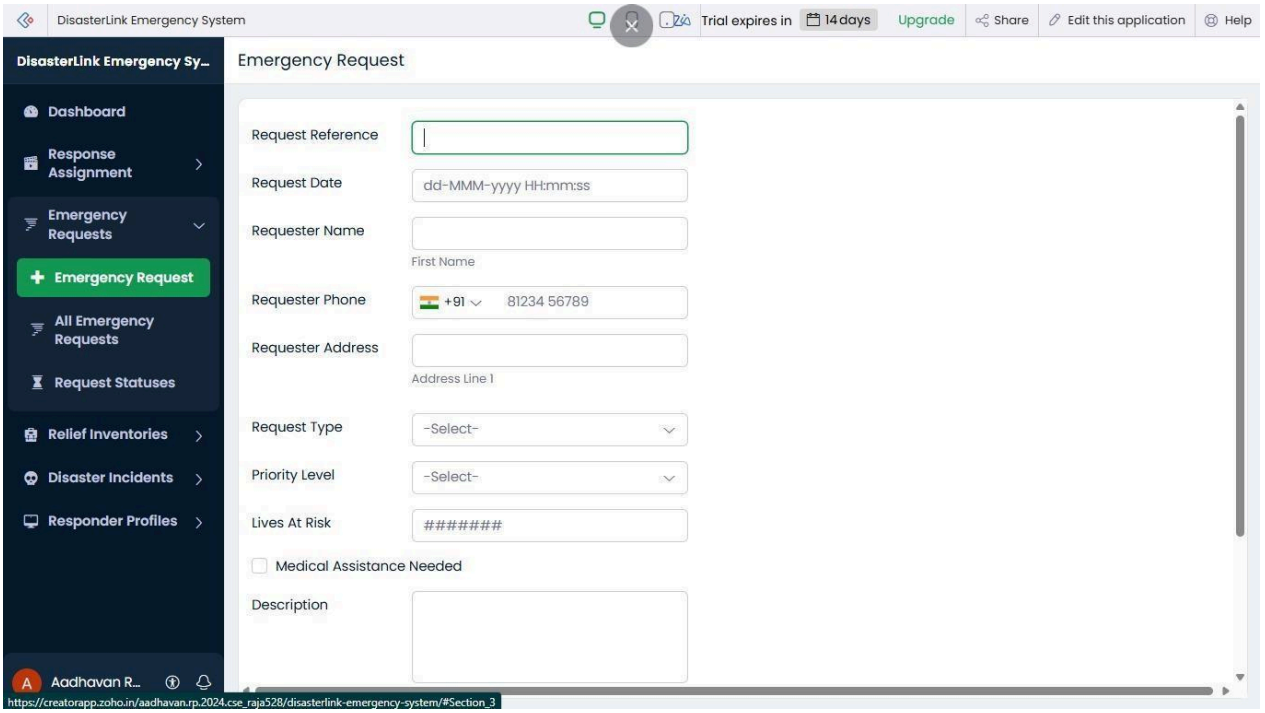


Figure 5.4: Emergency Request form – requester details, request type, priority and lives at risk.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should cover form validation, lookup relationships, inventory calculations, assignment creation, status updates, report visibility and role permissions. The test cases below are proposed verification cases. Actual result and final status should be completed after running each test in the live Zoho Creator application.

6.2 Test Cases

TC ID	Test Description	Input / Steps	Expected Result	Status
TC-01	Create emergency request	Enter valid requester details and request type.	Request record is saved with a reference.	To be verified
TC-02	Required-field validation	Submit request without mandatory fields.	System prompts user to complete required fields.	To be verified
TC-03	Set request priority	Select High / Medium / Low / Critical.	Selected priority is stored and visible in reports.	To be verified
TC-04	Create response assignment	Enter assignment, responder and destination details.	Assignment record is saved.	To be verified
TC-05	Reject zero allocation	Enter allocated quantity as zero.	System rejects the allocation.	To be verified
TC-06	Reject quantity above stock	Enter allocation greater than available quantity.	System rejects the allocation and stock remains unchanged.	To be verified
TC-07	Update inventory quantities	Allocate a valid quantity.	Available quantity decreases and reserved quantity increases by the same amount.	To be verified
TC-08	Update request status	Complete the response assignment workflow.	Linked emergency request status changes to Assigned.	To be verified
TC-09	Save delivery information	Enter dispatch time, delivery time and completion notes.	Delivery information is stored in the assignment.	To be verified
TC-10	View assignment report	Open Response Assignment Report.	Recorded assignment fields are displayed.	To be verified
TC-11	Dashboard summary	Open dashboard after adding records.	Cards and charts reflect current application data.	To be verified
TC-12	Role permission check	Login using each configured role.	Users can access only permitted records and actions.	To be verified

6.3 Debugging Deluge Scripts

Deluge scripts should be checked for correct form and field link names, data types, lookup access and workflow execution order. Debugging should include boundary values such as zero quantity, negative quantity, exact available quantity and a quantity greater than stock. Related inventory updates should be verified after each test.

6.4 Defect Log

Defect ID	Description	Severity	Resolution / Status
D-01	To be recorded during	—	Pending test execution

	application testing		
D-02	To be recorded during application testing	—	Pending test execution

6.5 User Acceptance Testing

User acceptance testing should be performed by an administrator or relief coordinator using realistic but non-sensitive sample records. The reviewer should confirm that requests can be recorded, stock can be allocated within limits, assignments can be reviewed and delivery information can be retained.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Deployment Strategy

The application should be tested in its development environment before it is published for operational use. Forms, lookup relationships, reports, workflows, Blueprint transitions and role permissions should be reviewed. The deployment URL and deployment date should be entered only after publishing and verification.

Deployment Item	Verification
Application access	Confirm authorized users can sign in.
Forms and fields	Check required fields, field types and lookup relationships.
Workflows	Test inventory validation and quantity updates.
Blueprint	Verify request review and assignment transitions.
Reports	Confirm report filters and displayed fields.
Dashboard	Confirm summary cards and charts match stored records.
Security	Test role and record-level access.
Backup / recovery	Review available Creator backup and export options.

7.2 Performance Optimisation

Reports should use relevant filters and avoid unnecessary data retrieval. Dashboard components should focus on operational measures such as pending requests, assignment counts, priority distribution and inventory availability.

7.3 Security Features

- Use authenticated application access and strong account credentials.
- Assign permissions according to operational responsibilities.
- Restrict requester contact details and other personal information to authorized users.
- Avoid storing passwords or sensitive access tokens in report documents.
- Review access to delivery evidence URLs and uploaded files.
- Maintain an audit trail for important assignment and inventory changes.

7.4 Governance and Administration

The application owner should maintain consistent naming for forms, fields and workflows. Changes to inventory logic, status transitions and user permissions should be tested before being applied to the live application. Sample data used for demonstrations should not expose real emergency requester information.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output Observed

The supplied dashboard screenshot shows the DisasterLink dashboard with the following displayed summary values: 6 response assignments, 4 emergency requests, 5 relief inventories and 1 disaster incident. The priority panel shows 2 High, 1 Low, 1 Medium and 0 Critical items. The page also includes a reserved quantity versus available quantity plot, a category-wise available quantity chart and quick links for creating records.

The Response Assignment form screenshot confirms the presence of fields for assignment reference, assignment date, assigned responder, responder phone, destination address, dispatch time, delivery time, delivery evidence URL and completion notes. The Response Assignment Report screenshot displays assignment rows with references REF-2026-001 to REF-2026-005 and corresponding responder and destination information. The Emergency Request form screenshot shows requester contact details, address, request type, priority, lives at risk, medical assistance and description fields.

8.2 Achievement of Objectives

Objective	Evidence / Outcome
Centralized incident and request records	Dashboard and Emergency Request form are shown in supplied screenshots.
Relief inventory tracking	Dashboard includes inventory count, quantity chart and category chart.
Responder and assignment coordination	Response Assignment form and report are shown.
Operational monitoring	Dashboard summary cards and priority indicators are visible.
Workflow automation	Project description specifies quantity validation, available/reserved stock updates and request status update; final runtime testing should be documented.

8.3 Comparison with Existing System

Aspect	Manual Process	DisasterLink
Request tracking	Paper, calls or separate spreadsheets	Centralized Emergency Request records
Inventory	Manual stock registers	Available and reserved quantity tracking
Assignments	Informal communication	Structured Response Assignment form and report
Priority	May be difficult to consolidate	Priority field and dashboard counts
Monitoring	Manual summary preparation	Dashboard cards, charts and reports

8.4 Limitations

The application's usefulness depends on accurate data entry, correct configuration of lookup relationships and reliable workflow execution. The supplied screenshots do not establish the presence of external SMS, map-routing, predictive analytics or government-system integrations. The application also depends on Zoho Creator availability and internet connectivity.

8.5 Low-Code Reflection

The project demonstrates the low-code approach through visual form creation, reports, dashboard components and workflow configuration. Deluge provides additional control for validation and inventory calculations.
More complex integrations or specialized analytics may require APIs or pro-code extensions.

8.6 Implementation Evidence

The screenshots included in Chapter 5 are the visual evidence supplied for the application. Dashboard values and sample report rows are reproduced as visible in those screenshots. Test-case outcomes are intentionally marked for verification rather than being represented as passed without execution evidence.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

DisasterLink – Intelligent Disaster Relief and Resource Coordination Platform provides a centralized low-code application model for managing disaster incidents, emergency requests, relief inventories, responder profiles and response assignments. The dashboard, forms and reports shown in the supplied screenshots demonstrate the main application interface and its operational information. The project description also defines workflow automation for validating allocations, updating available and reserved quantities and changing the linked request status.

By bringing these records together, DisasterLink aims to reduce scattered information, improve visibility into resource availability and support more organized emergency response coordination. The project demonstrates the use of Zoho Creator forms, reports, dashboards, lookup relationships, Blueprint and Deluge scripting in a practical application.

9.2 Future Enhancements

- Integrate map services to visualize incidents, destinations and responder locations.
- Add SMS, email or push notifications for urgent requests and assignment updates.
- Provide a citizen-facing portal for submitting and tracking emergency requests.
- Introduce role-based mobile views for responders in the field.
- Add predictive analytics to estimate relief demand using historical incident data.
- Integrate with approved government and emergency-service platforms through secure APIs.
- Provide audit dashboards and exportable operational reports.

9.3 Learning Outcomes and Career Relevance

The project provides practical exposure to low-code application development, data modelling, form design, lookup relationships, report creation, dashboard design, workflow automation, Deluge scripting, validation and testing. These concepts are relevant to business application development, process automation and software quality assurance.

REFERENCES

- Zoho Corporation, Zoho Creator Help Documentation. https://www.zoho.com/creator/help/
- Zoho Corporation, Deluge Scripting Documentation. https://www.zoho.com/deluge/help/
- Zoho Corporation, Zoho Creator – Forms, Reports, Pages and Workflows documentation.
- Course material: CB23F35 – Low-Code Application Development using Zoho Creator.
- DisasterLink Emergency System project description and application screenshots supplied for this report.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Feature	Purpose in DisasterLink	Evidence / Verification
Forms	Capture incident, request, inventory, responder and assignment records.	Screenshots supplied for Emergency Request and Response Assignment.
Lookup Fields	Connect requests, incidents, inventory and responders.	Verify configured lookup fields in Creator.
Reports	Display assignment and module records.	Response Assignment Report screenshot supplied.
Dashboard	Summarize assignments, requests, inventory and incidents.	Dashboard screenshot supplied.
Deluge	Validate allocation quantity and update inventory.	Verify scripts in live application.
Blueprint	Support request approval and assignment workflow.	Verify configured stages and transitions.

APPENDIX B – SAMPLE INVENTORY UPDATE

The project description provides the following demonstration: five Water Filters are allocated through a response assignment. The available quantity changes from 250 to 245, while the reserved quantity increases from 50 to 55.

Inventory Measure	Before Allocation	Allocated	After Allocation
Available Quantity	250	−5	245
Reserved Quantity	50	+5	55

This example illustrates the intended inventory update rule: Available Quantity = Previous Available Quantity
− Allocated Quantity; Reserved Quantity = Previous Reserved Quantity + Allocated Quantity.

APPENDIX C – USER MANUAL

C.1 Accessing the Application

Open the published DisasterLink Emergency System URL using a supported browser and sign in with an authorized Zoho Creator account. The application owner should provide access through the configured user-management process.

C.2 Viewing the Dashboard

Select Dashboard from the left navigation. Review the response assignment, emergency request, relief inventory and disaster incident summary cards. Use the priority panel and inventory charts to understand the current operational view.

C.3 Creating an Emergency Request

Open Emergency Requests and select the Emergency Request form. Enter the request reference and date, requester name and phone, address, request type, priority, lives at risk and medical assistance requirement. Enter a clear description and submit the record.

C.4 Creating a Response Assignment

Open Response Assignment. Enter the assignment reference and date, select or enter the assigned responder, provide the responder phone and destination address, and enter dispatch and delivery details when available. Add a delivery evidence URL and completion notes where applicable.

C.5 Reviewing Assignment Reports

Open Response Assignment Report to review assignment references, dates, responders, phone numbers, destination addresses and dispatch times. Use available report filters to locate a particular assignment.

APPENDIX D – PROJECT DETAILS TO BE COMPLETED BEFORE SUBMISSION

- Enter the student's register number on the cover page and Bonafide Certificate.
- Confirm the course code, academic year, department and guide/faculty details with the department.
- Add the published application URL only after deployment and verification.
- Verify the exact field link names and workflow stages against the live Zoho Creator application.
- Execute the test cases and replace 'To be verified' with actual results and evidence.
- Confirm whether any additional modules, integrations or role permissions are configured before describing them as implemented.